

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section.

FR-DLT-001: Load Customer and Order Data to Delta Tables

Title: Load Customer and Order Data to Delta Tables

Description: The system must read source CSV data from volume and load it into delta tables for customer and order data.

Preconditions:

- Customer data CSV file exists at `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata`
- Order data CSV file exists at `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

Main Flow / Functional Steps:

- Read customer data from CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into a delta table named "customer".
- Read order data from CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into a delta table named "order".

FR-DLT-002: Transform Order Data by Adding TotalAmount Column

Title: Transform Order Data by Adding TotalAmount Column

Description: The system must add a new column "TotalAmount" to the order delta table, which is the product of "PricePerUnit" and "Qty" columns.

Preconditions:

- Order delta table exists with columns "OrderId", "ItemName", "PricePerUnit", "Qty", "Date", and "CustId".

Main Flow / Functional Steps:

- Add a new column "TotalAmount" to the order delta table.
- Calculate "TotalAmount" as the product of "PricePerUnit" and "Qty" columns.

FR-DLT-003: Clean Customer and Order Data

Title: Clean Customer and Order Data

Description: The system must remove null and duplicate records from both customer and order delta tables.

Preconditions:

- Customer and order delta tables exist.

Main Flow / Functional Steps:

- Remove null records from customer delta table.
- Remove duplicate records from customer delta table.
- Remove null records from order delta table.
- Remove duplicate records from order delta table.

FR-DLT-004: Create ordersummary Table and Load Joined Data

Title: Create ordersummary Table and Load Joined Data

Description: The system must create an "ordersummary" table if it does not exist and load joined data from customer and order delta tables using "CustId" field.

Preconditions:

- Customer and order delta tables exist and are clean.
- Catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" exist.

Main Flow / Functional Steps:

- Create "ordersummary" table if it does not exist in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- Join customer and order delta tables using "CustId" field.
- Load joined data into "ordersummary" table as an SCD type 2 table.

FR-DLT-005: Update ordersummary Table on Customer Data Change

Title: Update ordersummary Table on Customer Data Change

Description: The system must update the "ordersummary" SCD type 2 table whenever there is a change in the customer table.

Preconditions:

- Customer delta table exists and is updated.
- "ordersummary" table exists as an SCD type 2 table.

Main Flow / Functional Steps:

- Detect changes in customer delta table.

- Update "ordersummary" table by making old records inactive and new records active.
- Update "StartDate" and "EndDate" columns accordingly to maintain history.

FR-DLT-006: Create customeraggregatespend Table and Load Aggregated Data

Title: Create customeraggregatespend Table and Load Aggregated Data

Description: The system must create a "customeraggregatespend" table if it does not exist and load aggregated data from "ordersummary" table.

Preconditions:

- "ordersummary" table exists.
- Catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" exist.

Main Flow / Functional Steps:

- Create "customeraggregatespend" table if it does not exist in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- Aggregate "TotalAmount" column from "ordersummary" table by "Name" and "Date" columns.
- Load aggregated data into "customeraggregatespend" table.

FR-DLT-007: Implement Delta Live Tables

Title: Implement Delta Live Tables

Description: The system must implement delta live tables for the above-mentioned functionality.

Preconditions:

- All previous functional requirements are met.

Main Flow / Functional Steps:

- Implement delta live tables for loading customer and order data.
- Implement delta live tables for transforming order data.
- Implement delta live tables for cleaning customer and order data.
- Implement delta live tables for creating and loading "ordersummary" table.
- Implement delta live tables for updating "ordersummary" table on customer data change.
- Implement delta live tables for creating and loading "customeraggregatespend" table.